

Long-term exposure to sodium nitrite and risk of esophageal carcinoma: a cohort study for 30 years.



The objective of this study is to investigate the risk of esophageal carcinoma in a cohort with long-term occupational exposure to sodium nitrite. The method used was a retrospective cohort study. A small wood screw manufacturer was founded in 1977 and closed down in 2000. In their production process, the sodium nitrite solution was used to serve as anticorrosive and coolant fluid. One hundred sixty workers in turning and milling shops had direct exposure to sodium nitrite through skin, mouth, and airway because of lack of occupational protective knowledge (study group), whereas 255 workers from other workshops without direct contact with sodium nitrite served as control group. The incidence, diagnosis, and treatment of esophageal carcinoma as well as other malignant tumors in these two groups were followed until the end of 2007. The sodium nitrite exposure time in the study group ranged from 16 to 23 years, with an average of 22.1 years. During 30 years of follow-up, there were 11 esophageal carcinomas and 10 other malignant tumors (4 hepatic cell carcinomas, 3 lung cancers, 2 breast cancers, and 1 leukemia) documented in the study group, while no cancer developed in the control group. The risk for esophageal carcinoma was significantly increased in the study group compared with the control group (relative risk = 1.26, 95% confidence interval = 1.08-1.46, chi-square = 116.83, $P < 0.001$). Long-term exposure to sodium nitrite markedly increases the risk of esophageal carcinoma in human body. © 2010 Copyright the Authors. Journal compilation © 2010, Wiley Periodicals, Inc. and the International Society for Diseases of the Esophagus.